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Towards Understanding Waiting in Video Games

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Abstract

Waiting is an everyday activity that is often present in video games. Waiting situations in games can happen during, for instance, loading screens, turn-based action in multiplayer games, and cut-scenes. Different waiting situations can cause both positive emotions, such as anticipation, and negative emotions, such as frustration. Thus, understanding how waiting relates to players' emotions can be beneficial when designing player experiences. However, academic discussion on waiting in games is quite scattered, and there is no comprehensive overview available on the subject. In this work-in-progress paper, we contribute a literature review on the topic of waiting in video games. Based on the results, five categories of academic discussion are identified. These are (1) models of time in video games, (2) games with waiting as a core game mechanic, (3) mathematical analysis of game situations, (4) games as a research tool, and (5) other perspectives. We also identify many waiting situations that are not discussed in the existing literature; hence, we hope that our work stimulates and informs future work on this underexplored but important topic.

CCS Concepts

• **Applied computing** → **Computer games**; • **Human-centered computing** → *HCI theory, concepts and models*.

Keywords

Video games, Player experience, Waiting

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1 Introduction

Waiting is a common, daily activity for most people. Whether it is waiting for the bus in the morning, waiting in line at the grocery store, or waiting for the coffee maker to finish brewing one's coffee, pauses in activity are present everywhere. Waiting situations can often create emotional responses in us, such as the frustration felt when the bus is not coming when it's supposed to, or the anticipation created by watching the coffee drip and knowing that soon it will be the time to drink it.

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In the domain of video games, much attention is paid to how a player feels in certain game situations. Various methods, such as pacing, difficulty, and inclusion of audio, visual, and narrative elements are used to nudge the player's emotions one way or another. It seems reasonable to assume that waiting could be used as one of such tools, as it has potential to create strong emotions in us.

Waiting, in the context of video games, could be described as a player experiencing a situation in which a certain amount of real time has to pass before they can continue to meaningfully interact with a game. While the concept of "meaningful interaction" is quite imprecise, this description highlights the subjectivity of how people experience waiting situations. As an example, watching a given cut-scene might be an interesting and valuable experience for one player, while another might consider it an annoyance they are required to sit through before they can proceed to the parts of the game that they find interesting.

Contribution. In this work-in-progress article, we contribute a literature review of how waiting has so far been discussed in the domain of game research. We additionally discuss some topics related to waiting that are not present in the current literature. To address this gap in follow-up work, we outline a plan for a questionnaire study. Our goal is to give new perspectives into how waiting can be utilized in video games, what types of waiting situations are common and most discussed in the domain of games, and what reactions and emotions these waiting situations can evoke in players. Furthermore, the literature review presents a categorized list of resources for further reading on the subject.

Research questions and scope. We articulate the research questions for our literature review as follows:

- RQ1: How much academic research has there been on the topic of waiting in video games? What are the main aspects and viewpoints of such discussion?
- RQ2: Have both positive and negative aspects of waiting in games been identified or discussed in literature?

As the goal of the literature review is to map out a rather wide area of discussion, the concept of waiting was not given a very strict definition when choosing which papers to include. Furthermore, the waiting situations that are considered relevant for this review can vary, as long as they involve the player having to wait for something to happen inside a video game. This includes both cases where the player waits with the game open and running, and cases where the player can close the game and perform other activities during the wait.

Although the goal of this work is to investigate waiting that happens inside video games, it's relevant to note that there have also been attempts to affect people's perception of wait times in other contexts by using games as a tool or a distraction. For instance,

Zambetta et al. [37] present their work to utilize an augmented reality game to lessen the perceived wait times in theme park queues. These types of projects were not considered to be within the scope of this review, as the waiting situation is not included in the game itself.

2 Background

> "Waiting is a stationary, yet dynamic, and unspecified time-frame phenomenon in which manifestations of uncertainty regarding personal outcomes remain in suspension for a limited time, but for the definite purpose of something expected." [9]

Waiting as a Human Experience. Waiting is a rather paradoxical human activity, as it does not necessarily require any activity at all. As Irvin's definition above shows, it can be quite difficult to find a specific, unambiguous definition for this phenomenon. Waiting situations can vary from the mundane experience of brewing coffee to the extreme stress of having to wait for medical procedures or news of life or death, so finding a unifying description feels difficult. Still, there is something that connects these experiences and makes us recognize them as manifestations of the same phenomenon. It is out of the scope of this work to explore the concept of waiting to its full extent, but this section will provide a brief overview of this discussion from various viewpoints.

In her concept analysis on waiting, Irvin [9] presents the concept from the point of view of healthcare. She identifies two critical attributes of waiting: uncertainty and loss of control. On the other hand, Dobler [4], whose focus is on social sciences, presents the following core criteria for waiting: future orientation, passivity, uncertainty, stasis, and absence of intrinsic value. As can be seen, both authors identify uncertainty as a key concept when discussing waiting. Dobler further specifies that uncertainty does not necessarily mean that one is uncertain if an event will happen, but it might not be known when it will occur, or how long the wait time will *feel*.

Discussion on waiting often revolves around its subjectivity. Much research has been done on how waiting affects people's emotions and behaviour in different scenarios, such as when seeking asylum [21] or in everyday customer experiences [14]. Although people can react to waiting situations in varying ways, waiting is most often associated with negative emotional responses, such as frustration, annoyance, uncertainty, anxiety or anger [29]. However, the one emotional response that it is perhaps most often associated with is boredom. Zakay [36] argues that boredom is an opposite to the so-called flow state during which the perception of time fades away. On the other hand, when a person is experiencing boredom, the perceived wait time becomes longer. This means that based on how a person is feeling, a waiting situation can either feel longer or shorter than it is in reality.

Although waiting is commonly seen as a negative experience and much research has been done to come up with methods to lessen the perceived wait times in various areas, there can also be positive aspects to waiting and waiting situations. Ryan et al. [23] attempt to outline positive aspects to waiting in the context of consumers who are waiting for a service. According to the authors, waiting for something increases its perceived value and encourages positive anticipation. Indeed, waiting for a good thing can feel exciting and motivating in other contexts as well. Additionally, Dobler [4] notes that people's capability to wait is often what allows for cooperation: if no party was willing to wait for the other, everyone had to be exact when they arrived and working together would become difficult.

Waiting in HCI. In the domain of Human-Computer Interaction (HCI), negative waiting experiences are recognized as a common problem, e.g., in web services; hence, wait times on the Internet have received a lot of discussion. Some topics of discussion include the experiences of frustration and powerlessness in the context of slow internet speeds [19], internet users' willingness to wait for web pages to load [20, 28], and customer experiences when using a virtual queuing system [7].

In addition to experiences of waiting online, visualisations of wait times are a common topic in HCI research. Lallemand and Gronier [15], Chen and Li [2], and Hohenstein et al. [6] discuss how the visualisation of loading progress in waiting situations—e.g., loading screens or loading bars—affects the users' perception and satisfaction of using a system. The results of Lallemand and Gronier [15] in particular show that people's perception of wait times and the subsequent emotional responses to the waiting situation can be affected by how the wait is visualised. A lack of indication for the remaining wait time causes frustration, but too much information during waiting can make the wait feel longer.

Why Should We Study Waiting in Games? While the findings from HCI research can be useful for game design and analysis, we believe that both researchers and practitioners can benefit from research on waiting experiences in games specifically. In HCI, the discussion on waiting centers largely on how to limit boredom and frustration caused by waiting, but games do not necessarily benefit from only this approach. In fact, players' frustration can sometimes be a desirable effect in games, as the emotion can e.g. motivate the player to explore different strategies or complement the game's narrative purposes. Moreover, design of game situations that create more complex emotions, such as dread, anticipation, or hope, could benefit from a better understanding of how players experience waiting in games.

3 Method

The literature review was conducted based on guidelines presented by Snyder [24]. A semi-systematic review was chosen as the best approach, as Snyder notes this to be a good approach for providing overviews, mapping themes, and identifying knowledge gaps in literature, which aligns with our goals.

The process during the review was as follows:

(1) Design search queries based on the research questions
(2) Gather articles using an academic search engine and the designed queries
(3) Read and analyse the articles, discarding irrelevant papers
(4) Identify common themes and divide the articles into categories

The review was done by collecting and analysing search results from a search tool focused on academic literature. We initially tested Google Scholar, Semantic Scholar, and Scopus with test queries such as "waiting AND video game*" and chose Google Scholar as the

tool for the review as it produced most relevant search results. The four queries that were used in the review are listed in table 1. To keep the scope of the review suitable, the queries were kept quite simple and focused on a small number of search terms.

For each of the four queries, the first 50 results were assessed for relevancy based on the research questions and scope that were outlined in Section 1. Only papers written in English were included in the review. During the search process, the article's title, keywords, publication journal, and abstract were considered, and relevant articles were saved for further analysis.

After the articles were collected, they were read and analysed more thoroughly. During this process, some of the papers that initially seemed relevant were discarded, as they were found to not correspond to the goal of the review. The results of the analysis are presented in the next section.

4 Results

In total, 21 relevant articles were found during the review. Table 1 shows the total numbers of relevant results for each search query, as well as how they were divided into five categories which will be further explained in the following sections. Papers which discussed multiple of the five categories are marked to the table under the category that was deemed most central to the paper's discussion. Many of the articles appeared in the search results of several queries, in which case they are included in multiple rows of the table.

Several common threads can be identified in the found articles. Based on these similarities, the papers were divided into categories, which are (1) models of time in video games, (2) games with waiting as a core game mechanic, (3) mathematical analysis of game situations, and (4) games as a research tool. Additionally, some of the papers that were found could not be placed in any common category, so they were grouped under category (5) other perspectives. The following sections discuss each category and give an overview of the related papers.

4.1 Models of time in video games

Expectedly, many articles referencing the topic of waiting in games discuss different ways to model and perceive time in video games. Many of these papers are not specifically focused on waiting as the main phenomenon, but waiting situations are often referenced in passing when discussing different aspects of time and timing in video games. Overall, these works provide useful models and terminology for discussing time, which can be beneficial for understanding the concept of waiting as well.

Zagal and Mateas [35] present a survey on time and temporality in video games. The authors present the concept of a "temporal frame" and discuss four different frames which can be used to analyse time in games. These are "real-world time", "gameworld time", "coordination time", and "fictive time". Each of these frames can be used to analyse time from a different perspective. For example, real-world time refers to the physical time in the world outside the game, whereas gameworld time refers to how time is presented in the game world – a full day from sunrise to sunset in gameworld time might only take ten minutes in real-world time. As many games allow time to be slowed down, sped up, or paused, and players can re-load old save files, models of time in games can become quite complex.

Although the terminology is often different, many articles that were discovered refer to similar categorizations of time as [35]. Van Meurs [25] presents several such models. Relevant to the discussion on waiting, some of these models include times when the player is not able to interact with the game, such as loading screens and cut-scenes, whereas others only consider the interactive periods of playing. Van Meurs additionally presents the concept of "dead time", which refers to time spent playing without interesting or challenging interaction. Examples of dead time are activities known as "farming" or "grinding" where players take repetitive actions in order to gain resources or a reward, but waiting situations can be thought of as dead time as well.

Vanderhoef and Payne [26] also discuss different models and representations of game time. They consider how these representations can be linked to the more general cultural ideas of labour and productivity in the current economic systems in the Western world. In their article *Press X to wait: The cultural politics of slow game time in Red Dead Redemption 2*, Vanderhoef and Payne present a discourse analysis on how the slow pace of action, something they refer to as "slow game time", was perceived and experienced by players of Red Dead Redemption 2 [G2]. Furthermore, the authors present a case for the benefits of slow pacing and waiting in the game, as they complement the game's "focus on journey over destination and process over productivity". This is the only article that was found which addresses the potential of using waiting situations to emphasize a game's narrative decisions.

4.2 Games with waiting as a core mechanic

Many of the discovered papers focus on discussing games and game genres where waiting is used as a core game mechanic. In the publications found, waiting as a mechanic was mostly discussed in the context of idle games, social network games, and a subsection of mobile games. While the papers discuss these games from different points of view, themes of the value of players' time and attitudes toward casual games are common.

Idle games are games where the player is often encouraged to leave the game running in the background for periods of time without interaction. In their paper *Playing to wait: A taxonomy of idle games*, Alharthi et al. [1] analyse mechanics, characteristics, and interfaces of various idle games and present a taxonomy for categorizing them. They suggest that one of idle games' unique aspects is that they center and value waiting as a part of the gameplay, and furthermore that these waiting periods can be beneficial as they give players time to set play aside and think about their future choices in the game.

Different types of waiting mechanics are common especially in the mobile game sphere. Keogh and Richardson [11] discuss so-called "background games" (which they also refer to as "ambient games") in the context of mobile gaming. These are games where players play short periods at a time and the gameplay generally consists of setting up time-consuming processes that the game automatically completes in the background while the player is focusing on other activities. The player can then later return to the game to collect rewards from finished processes and set up new

Table 1: Articles that were found for each search query

Query	Models of time	Waiting as a core mechanic	Mathematical analysis	Games as a research tool	Other perspectives	Total
waiting AND game*	-	[1]	[5, 18]	-	[3]	4
waiting AND video game*	[26]	[11]	-	[8, 22, 27, 30–34]	[3, 12]	12
waiting AND gaming	[26, 35]	[1, 11, 25]	[5, 16, 18]	[13]	[10]	10
wait* AND video game*	[26]	[1, 11, 25]	-	[22, 30]	[3, 12, 17]	9

ones to begin. The authors contrast this model to many "traditional social games" where the player must wait for an energy bar of some kind to refill before continuing to progress in the game. In background games, the players are "not waiting to perform tasks but waiting for set tasks to be completed".

Similar to the concept of background games, van Meurs [25] discusses social network games that became popular on social media sites, such as Facebook. A central aspect of social network games is social play. As van Meurs presents, these games are "asynchronous multi-player experiences", where players interact with each other without the need to be online at the same time. This multiplayer element brings a new aspect of waiting, as in these games people are not waiting for game processes to complete, they are instead waiting for other people to continue playing and take certain actions.

Both [11] and [25] additionally discuss some valid criticisms of waiting as a game mechanic. Keogh and Richardson [11] bring up the concept of "pay to not play" in the context of waiting in background games. It is common that many of these games allow players to skip waiting periods by spending in-game currency that can be obtained with real money, which can put players in unequal positions based on their financial situations. However, in their survey, Keogh and Richardson note that many players expressed that they gain gratification from choosing to wait instead of paying to progress in the game. Similarly, the popularity of idle games presented by Alharthi et al. [1] suggests that waiting has potential to be a rewarding game mechanic for many players.

4.3 Mathematical analysis of game situations

Some of the papers discuss the concept of waiting from a more mathematical perspective. Both [18] and [5] discuss the expected waiting time in quite simple games: rock-paper-scissors and a roulette respectively. In these discussions, "waiting time" does not refer to the real-world time it takes to finish a game, but rather the expected number of game rounds before the game ends with a given game rule set and configuration. On the other hand, [16] discusses how the inclusion of "pass" moves affects the results in combinatorial scoring games.

4.4 Games as a research tool

Many of the articles that were found in the review did not aim to contribute to games research. Rather, they analysed people's behaviour in waiting situations from the perspective of psychology and used games as a tool to simulate these waiting situations. The results of these analyses can still be relevant to the discussion of waiting in games, however, as they provide insight into how players evaluate waiting situations and decide when it's beneficial for them to wait.

Igarzábal et al. [8] present an experiment where the participants were asked to wait in a waiting room in virtual reality. They discuss the connection between boredom and the perceived passage of time and compare people's experiences of waiting situations in virtual and physical waiting rooms. Interestingly, the authors find that people find waiting in a virtual space more tedious compared to a similar waiting space in the physical world. Similarly, Kino et al. [13] analyse people's waiting behaviour in a public square utilizing a game-like simulation of the space.

Many papers analysing decision-making in delayed gratification situations use games as a tool to present participants with different options to choose from. A common theme is to create a scenario where a participant can opt to wait for a certain amount of time in order to change (typically increase) the reward for some action. [32], [34], [27], [22], [31], and [33] all utilize first-person shooter games to create variations of such situations, whereas [30] opts for a simpler box-clicking game.

4.5 Other perspectives

In addition to the four categories discussed above, there were also relevant papers that we could not fit into any common category. These are briefly summarized below.

Cheng [3] presents an analysis on the different ways cut-scenes can be utilized in video games. Cheng specifically focuses on how cut-scenes can be turned from "passive" play where the player is not interacting with the game to a more "active" play experience. However, the work does not yet dive very deeply into how the passivity and waiting duration of cut-scenes are experienced by players.

King et al. [12] present a taxonomy for analysing video games from the point of view of game addiction. While the paper does not analyse waiting in games in much depth, the authors present a list of different waiting situations in games, which can be useful for further analysis of the subject.

Johnson and Wiles [10] present analysis on how video games commonly break Nielsen's guidelines for heuristic evaluation, and how these common missteps affect games' ability to induce the "flow state" in players. The authors discuss waiting situations, such as unskippable cut-scenes (referred to in the text as "movies") and waiting for game content to load. These violate the heuristic of flexibility and efficiency and are inherently frustrating to the player.

Finally, Liu et al. [17] present an experiment where players encounter simulated game loading situations of varying durations and

rate their quality of experience. Based on the experiment, the authors then present a model for predicting how load times in games affect players' quality of experience.

5 Discussion

Overall, waiting has not been discussed in the domain of video game research in much depth, as indicated by the quite small number of publications found in the literature review. Moreover, most of the articles that were found do not discuss waiting in games as a central topic. Rather, they focus on time perception or more general behavioural analysis, and only discuss waiting situations in passing.

When it comes to articles that focus on waiting situations during gameplay, nearly all the discussion centers on idle, background, and social network games. Apart from one case analysis on *Red Dead Redemption 2* [26], this leaves waiting in other game genres unexplored. Some discussion also centers around common waiting situations in games, such as loading screens and cut-scenes, but these topics are mostly covered by individual papers so there is limited discourse related to them.

In the articles that were found in the literature review, both positive and negative effects of waiting were discussed. Similar to the discussion outside of video games, mentions of emotions such as frustration and annoyance were reoccurring. On the other hand, the beneficial aspects of waiting that were mentioned include creating space for anticipation and giving players more time to think over their future actions in the game.

There are many common aspects of waiting in games that were only briefly discussed in the literature and could be explored in more depth in the future. Inclusion of various kinds of waiting situations affects how players respond to and interact with a game, so further analysis on players' waiting experiences could be beneficial for understanding and improving players' overall experiences while playing. In our opinion, at least the following aspects seem particularly interesting for further analysis: 1) Waiting as an introspective activity, 2) Waiting as a social activity, and 3) The use of waiting to create a sense of suspense and motivate the player to explore different aspects of the game.

Waiting as an Introspective Activity. Many games contain moments where the player has to, e.g., wait for the weather to change, recover from exhaustion, or travel to a new location. These quiet moments could be seen as giving the player time for introspection, as the game does not demand their full focus during the waiting period. On the other hand, these moments could be experienced as frustrating or boring by some players, so many games offer options to skip the wait, such as by letting the player teleport to the desired location instead of travelling there in real time. Such "fast travel" is a common mechanic especially in open world games, but according to player discussions on social media such as Reddit, there are games that manage to make regular travelling so interesting or beautiful that players may choose to abandon fast travel, e.g., *"I dont fast travel… because I purely don't want to, the joy of riding around NC, listening to radio station, getting txts, it's just… so well done."* [1], *"Free falling off a skyscraper, gaining speed and then web swinging at the last moment to slingshot yourself never got old for me."*[2]

Waiting as a Social Activity. Waiting can also create time for socialising. For instance, many match-based online multiplayer games contain waiting rooms where players can interact with each other before the game match begins. In cooperative multiplayer games, waiting situations can be used for planning and conversation, which might be difficult to do during the active parts of gameplay.

Creating Suspense and Motivating Exploration. Other underexplored aspects of waiting include its use in creating suspense and as motivation for exploration. A good example of this is the guest structure in the game Cyberpunk 2077 [G1]. In this game, missions tend to be broken down into multiple parts. Between these parts, the player has to wait to hear back from a relevant character to see how their mission has progressed. This creates a sense of suspense and anticipation for the player, as they have to wait to learn the results of their previous choices. Additionally, this structure encourages the player to explore other quests and missions while they are waiting. These mechanics give a clear example of how waiting situations can be used to affect player experience and behaviour.

6 Conclusion and Future Work

We have presented a literature review on the topic of waiting in video games. The papers found in the review were divided into five categories according to their perspective on waiting. Additionally, we have identified and discussed some underexplored perspectives that warrant future work.

In our future work, we plan to conduct a questionnaire survey aimed at game designers and players to gain more targeted insights into waiting in games. The survey will ask participants to recall and describe both well and badly designed cases of waiting they have encountered. We aim to conduct a thematic analysis on the data, and possibly build a taxonomy of different waiting situations in games. Our goal is to build a better understanding of the design space and the opportunities and challenges related to waiting in games, as well as give practical examples that other designers and researchers can learn from. In addition to custom questionnaires, it appears that social media discussions on topics such as fast travel in open world games may be useful data sources for details and examples of both good and bad design decisions.

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